

CL-FEP: An End-State Free Energy Perturbation Approach

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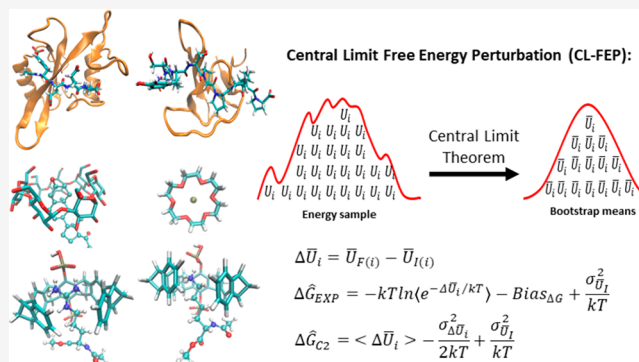


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ABSTRACT: The Central Limit Free Energy Perturbation (CL-FEP) approach, based on the Free Energy Perturbation (FEP) theory and the Central Limit (CL) theorem, allows evaluating the FEP identity directly from the energy samples of the end states of a system transformation. In CL-FEP, energies obtained from explicit solvent simulations are used to estimate the absolute free energy change. No fitted parameters are introduced in our implementation, and no stratification is needed to obtain accurate free energy evaluations. We first illustrate the applicability of CL-FEP in four dissimilar benchmark systems from host–guest to proteins–peptide complexes, for which the deviation from the experimental values was below 1 kcal/mol. Next, we extended the validation to three sets of complexes comprising 25 host–guest and protein–ligand systems. There, the mean absolute error of the free energy changes calculated with CL-FEP is between 1.1 and 1.4 kcal/mol among the different sets, which is in the range of accuracy of the most reliable free energy approaches applied to these systems. CL-FEP is an unbiased free energy change estimator that also profits from a bootstrapping algorithm, which makes the evaluation of convergence and confidence an inherent component of the procedure. Finally, we developed CLFEP.pl, an automatic tool to compute free energy changes directly from the energy output of standard molecular dynamics simulations.



1. INTRODUCTION

The change in free energy is a key thermodynamic parameter, which evaluation allows understanding chemical and biophysical processes taking place in nature at standard conditions. Thus, free energy calculations have found many applications and are nowadays an essential tool for the study of chemical and biological systems. Several computational approaches have been established to estimate free energy changes from molecular simulations.¹ The accuracy and general applicability of these methodologies vary significantly, with end-state approaches like linear interaction energy^{2,3} (LIE) and MM/(P/G)BSA,^{4–6} being among those with less computational cost. However, such feasibility results in diminished accuracy with respect to more elaborated and computationally demanding approaches like extended-system adaptive biasing force^{7,8} (eABF), multistate Bennett acceptance ratio⁹ (MBAR), stratified free energy perturbation^{10,11} (FEP), or thermodynamic integration (TI).^{10,12}

The lower accuracy and confidence of fast end-state methods with respect to more computationally demanding free energy approaches is related to simplifications introduced in the Hamiltonian of the systems (e.g., implicit solvent approaches) and in the free energy estimator itself. LIE, for instance, is a fitted linear estimator that correlates the free energy change with the difference in the nonbonded interactions between the initial and final states of the process.^{2,3} The MM/(P/G)BSA approach is a linear average estimator that evaluates the free energy change

from the average energy difference between the initial and final states.^{4,5} There, the considerable energetic noise that typically accompanies explicit solvent simulations is reduced by removing the solvent molecules and approaching solvation energies using implicit solvent electrostatics.^{13,14} Additionally, an empirical entropic contribution is usually added as a way to compensate the use of a nonexact estimator.^{15,16}

Free energy perturbation (FEP) theory¹¹ is recognized as the oldest exact estimator of free energy differences. FEP theory and other methods like the Bennett acceptance ratio¹⁷ are theoretically defined as free energy estimators based on energy samples from the end states of the system transformation. Despite the exactness of these formalisms, stratified sampling between the initial and final states is commonly needed to obtain accurate free energy change evaluations.¹ FEP calculations have increasingly benefited from advances in scientific computing.^{1,18–20} However, even with modern computers, the stratified approach is time-consuming enough to prevent the automated and massive use of free energy perturbation calculations, particularly in drug discovery and ligand screening processes.

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In FEP theory,^{10,11,21} the Hamiltonian of the system in the final state of a transformation is expressed as the sum of the Hamiltonian of the initial state and a perturbation term that accounts for the energetic difference between the two states. The free energy difference between the two states (initial and final) is then exactly evaluated as a function of the exponential average of the perturbation term over the initial state

$$\Delta G = -kT \ln \langle e^{-\Delta U/kT} \rangle_i \quad (1)$$

where k is the Boltzmann constant, T is the temperature, ΔG is the free energy, and $\Delta U = U_F - U_I$ corresponds to the variation of potential energy between the final and initial states. The FEP identity was initially derived for the canonical ensemble.^{1,11} However, the validity of this equality has been proved also in the NPT ensemble.^{22–24} Consequently, we use the Gibbs free energy to formulate the FEP equation.

Despite the straightforward relation stated by the FEP identity, a successful and direct use of eq 1 can only be achieved if the probability distribution ($P(\Delta U)$) is a very narrow function of ΔU . Such scenario is far from the energy sampling obtained in common molecular simulations within a temperature regime, where the total energy shows fluctuations in the order of hundreds of times the kT level. The huge variance of the energy samples is thus an issue for the practical use of the FEP identity. The traditional way to overcome the energy variance problem is the use of stratification.¹ This approach relies on sampling multiple intermediate states along the system transformation, with small and gradual changes of a given state with respect to the previous one. This way, $P(\Delta U_{i,i+1})$ for two consecutive states i and $i+1$ is sufficiently narrow to allow for the direct evaluation of the corresponding free energy difference, $\Delta G_{i,i+1}$. Then, the complete free energy difference can be recovered as

$$\Delta G = \sum_{i=1}^{N-1} \Delta G_{i,i+1} = -kT \sum_{i=1}^{N-1} \ln \langle e^{-\Delta U_{i,i+1}/kT} \rangle_i \quad (2)$$

The states generated along the stratification do not need to be physically relevant. In fact, the Hamiltonian can be considered as a function of an occupancy parameter, λ , in a way that $0 \leq \lambda \leq 1$, where $\lambda = 0$ corresponds to the initial state, and $\lambda = 1$ corresponds to the final state.

The stratified approach to evaluate the FEP identity is an inherent part of FEP calculations and relies on the fundamental argument that the free energy, as a state function, is independent of the path chosen. Even when stratification is undoubtedly a reliable approach to the FEP identity, accurate end-state implementations are also desirable.

To this end, we developed the Central Limit Free Energy Perturbation (CL-FEP) approach, which combines the FEP theory with the central limit theorem to reduce the energetic noise of the potential energy distribution ($P(\Delta U)$). The energy sampling in each state is obtained from explicit solvent simulations without modifications in the molecular mechanics Hamiltonian or in the molecular representation of the system. This way, CL-FEP allows for the exact evaluation of the exponential average (eq 1) estimator using the energies from the initial and final states of a system transformation without the need of stratification. We note that by exact we mean that the classical molecular and statistical mechanics formalisms are used without any *ad hoc* empirical modification or fitting procedures. By accurate we refer to how well the estimated free energy values reproduce experimental measurements. In this context, the binding free energies calculated using CL-FEP show mean

absolute errors below 1.5 kcal/mol from experimental values, which is in the range of the most reliable state-of-the-art free energy methods.⁷

2. THEORETICAL BACKGROUND

2.1. Bias, Variance, and Error of the Exponential Average Estimator. In this section we describe the bias, variance, and error associated with the exponential average estimator in the form of the Jarzynski equality.^{25,26} This estimator, also referred to as the nonequilibrium work (NEW) method,¹ is a generalized form of the original FEP identity, which depends on the exponential average of nonequilibrium energy differences (work values)^{25,26}

$$\Delta G = -kT \ln \langle e^{-W/kT} \rangle \quad (3)$$

where k is the Boltzmann constant, T is the temperature, and W represents measurements of nonequilibrium energy differences between the initial state and the final state of the system. Despite the exact nature of this free energy change estimator, its applicability is reduced when the energy fluctuations surpass several times the level of kT , which is the most common scenario even in equilibrium simulations.

It has been established that the Jarzynski estimator is biased for any finite sampling size,²⁷ which is a consequence of the convex nature of the exponential function according to the Jensen's inequality.^{28,29} This bias is a systematic error that increases with the variance of W , thus affecting the accuracy of the free energy estimate. The magnitude of the bias is generally unknown. For the near-equilibrium regime (regime in which the probability distribution of W is Gaussian but without restriction to the amplitude of its variance), a bias-corrected free energy estimator was proposed (eq 4)²⁷

$$\Delta \hat{G} = \Delta G - B_{\Delta G} = \langle W \rangle - \frac{\sigma_W^2}{2kT} \quad (4)$$

where $B_{\Delta G}$ represents the bias of the exponential average (EXP) estimator, and σ_W^2 is the variance of the work samples. The last expression of eq 4 is the second order cumulant expansion (C2) of the EXP estimator, which is exact and unbiased, provided the Gaussian distribution of the energy differences, i.e., near equilibrium regime. The C2 estimator has been used for the determination of free energies from steered^{22,30} and accelerated molecular dynamics simulations.³¹

The statistical parameters that characterize the performance of free energy estimators are the bias (B), the variance (σ^2), and the root mean square error (RMSE), which are defined as follows

$$B_{\Delta G} = \langle \Delta G_{i(N)} \rangle - \Delta G^0 \quad (5)$$

$$\sigma_{\Delta G}^2 = \langle (\Delta G_{i(N)} - \langle \Delta G_{i(N)} \rangle)^2 \rangle \quad (6)$$

$$\text{RMSE}_{\Delta G} = \sqrt{\sigma_{\Delta G}^2 + B_{\Delta G}^2} \quad (7)$$

where ΔG^0 is the experimental standard free energy change, and $\langle \Delta G_{i(N)} \rangle$ is the average free energy value among multiple independent estimations (O), each one made using N energy samples.

$$\langle \Delta G_{i(N)} \rangle = \lim_{O \rightarrow \infty} \frac{1}{O} \sum_{i=1}^O \Delta G_{i(N)} \quad (8)$$

Bustamante and co-workers analyzed the bias and error in the free energy change using Jarzynski's equality in two contexts: (i) a small sampling regime (rather low N values) and (ii) in the large N limit²⁷ and concluded that the N limit is usually unreachable in simulations. Thus, we focus on the formulations to estimate the bias and the variance in the small N regime which, provided the compliance of the near-equilibrium regime, can be written as²⁷

$$B_{\Delta G} \approx \frac{W_{dis}}{N^{\alpha_B}} \quad (9)$$

$$\sigma_{\Delta G}^2 \approx \frac{\sigma_W^2}{N^{\alpha_V}} \quad (10)$$

where W_{dis} is the so-called dissipated work, which can be evaluated as

$$W_{dis} = \langle W \rangle - \Delta G = \frac{\sigma_W^2}{2kT} \quad (11)$$

The alpha functions (α_B and α_V) are given by the general equation

$$\alpha = \frac{\ln[2CW_{dis}/kT]}{\ln[C(e^{2W_{dis}/kT} - 1)]} \quad (12)$$

where C is an empirical constant related to the boundary between the small sampling and the large N regimes with optimal values of $C = 15$ and $C = 50$ for the bias and the variance of the free energy, respectively (Bustamante et al.²⁷). Accordingly, a bias-corrected exponential average estimator can be defined as

$$\Delta \hat{G} = -kT \ln \langle e^{-W/kT} \rangle - B_{\Delta G} = \Delta G - B_{\Delta G} \quad (13)$$

where $B_{\Delta G}$ is the bias.

An additional first-order correction was also proposed:²⁷

$$\Delta \hat{G} = \Delta G - B_{\Delta G} - \frac{B_{\Delta G}}{N^{\alpha_B}} \quad (14)$$

However, this additional term delivered systematic improvements only in the cases of large dissipated work ($>10kT$).²⁷ Here, we corrected the exponential average estimator by subtracting the bias from the raw estimator outcome (eq 13). The additional first-order contribution of the bias is incorporated in the calculation of the RMSE (eq 7) in replacement of the original bias. Thus, the error of the bias-corrected estimator is written as

$$\text{RMSE}_{\Delta \hat{G}} = \sqrt{\sigma_{\Delta G}^2 + \left(\frac{B_{\Delta G}}{N^{\alpha_B}}\right)^2} \quad (15)$$

In practice, this error quantifies the precision of the final estimate. Using the formalism described above, the bias, variance, and error of the exponential average estimator can be evaluated for any sampling obeying the near-equilibrium regime and with a limited number, N , of energy samples. We note that eq 10 relates the variance of the free energy change to the variance of the energy difference values, σ_W^2 , and the size, N , of the sample.

An interesting point, however, is to analyze the statistical mechanics counterpart of this variance. It is known that the variance of the enthalpy and entropy is dependent on the heat capacity of the system according to^{32,33}

$$\sigma_{\Delta H}^2 = kT^2 \Delta C_p \quad (16)$$

$$\sigma_{\Delta S}^2 = k \Delta C_p \quad (17)$$

These expressions are a manifestation of the fluctuation–dissipation theorem.³² Nonetheless, Prabhu and Sharpa³³ concluded from a statistical mechanic analysis of the heat capacity that the entropy and enthalpy variances are canceled by the covariance of the enthalpy–entropy fluctuations, thus rendering a free energy change with null variance.³³

$$\sigma_{\Delta H \Delta S}^2 = kT \Delta C_p \quad (18)$$

$$\sigma_{\Delta G}^2 = \sigma_{\Delta H}^2 + T^2 \sigma_{\Delta S}^2 - 2T \sigma_{\Delta H \Delta S}^2 = 0 \quad (19)$$

Hence, the variance defined by eq 10 does not obey thermal fluctuations in real systems, but it can be seen as a standard error generated by the finite sampling used to make the free energy estimation. Next, we describe how we established a suitable threshold value, $(kT)^2$, for the variance of the free energy change estimations with any finite sampling approach (see below, eq 25). Subsequently, we use this fixed variance of the final free energy estimate to effectively modulate the original variance of the energy samples for each state, thus controlling the convergence of the CL-FEP estimator, see below.

We note that the compensation between the thermal variances of the enthalpy and entropy terms does not reflect the random errors associated with estimating their mean values from finite system samples. The standard error of the mean (SEM) is the random error obtained from estimating the mean values of multiple samples from the same population. The SEM for the average enthalpy or entropy is formulated as

$$\text{SEM}_X = \sqrt{\frac{\sigma_X^2}{N}} \quad (20)$$

where X is the enthalpy or the entropy, and N is a magnitude associated with the size of the sampling. The stochastic nature of this error prevents any covariance between the enthalpy and the entropic SEM values. Then, the so-called variance of the free energy estimator, eq 10, can be represented as a sum of the squared SEM values of the enthalpy and the entropy contributions

$$\sigma_{\Delta G}^2 = \text{SEM}_{\Delta H}^2 + T^2 \text{SEM}_{\Delta S}^2 \quad (21)$$

Next, we make use of eqs 16 and 17, by writing the heat capacity as

$$C_p = \frac{\Pi}{2} k \quad (22)$$

where Π is the accessible number of degrees of freedom, or microstates, of a given system sample. Since enthalpy (and entropy) variances are related to the fluctuations along the accessible microstates of the system, the SEM for the enthalpy and entropy can be estimated as

$$\text{SEM}_{\Delta H}^2 \approx \frac{\sigma_{\Delta H}^2}{\Pi} = \frac{1}{2} k^2 T^2 \quad (23)$$

$$\text{SEM}_{\Delta S}^2 \approx \frac{\sigma_{\Delta S}^2}{\Pi} = \frac{1}{2} k^2 \quad (24)$$

By combining eqs 23 and 24 with eqs 10 and 21, a fixed value for the expected variance of the free energy can be established for any finite sampling of the system

$$\sigma_{\Delta G}^2 = (kT)^2 \approx \frac{\sigma_W^2}{N^{\alpha_V}} \quad (25)$$

Thus, eq 25, which is valid in the near equilibrium regime, can be used to determine the convergence of free energy calculations by fixing a threshold value to the variance of the free energy change estimation.

2.2. Central Limit Free Energy Perturbation Approach (CL-FEP). The FEP estimator, together with the formulations described above, provides the theoretical framework to our Central Limit Free Energy Perturbation approach (CL-FEP). This approach aims to smooth, in a straightforward manner, the large energetic noise of simulations in explicit solvent, which typically prevents the direct application of the original FEP identity using the energies of the end states. The use of a smoothing technique to tailor the convergence of stratified FEP approaches was shown recently.³⁴

In CL-FEP, the energy samples of the initial and final states can be obtained from molecular dynamics or Monte Carlo simulations. However, these energies are usually plagued with a huge variance and do not necessarily follow a Gaussian distribution. Hence, our first aim is to generate a *new sample* of energy differences that satisfies the near-equilibrium regime while, at the same time, the energetic noise is reduced. That is

- i) $\langle \Delta U^* \rangle \approx \langle \Delta U \rangle$; $P(\Delta U^*)$ follows a Gaussian distribution.
- ii) $\sigma(\Delta U^*) < \sigma(\Delta U)$

where the star (*) represents the newly generated sample. The first of these conditions is due to the fact that the new sample must describe the same system transformation as the original one. Thus, the average energy difference between the final and initial states should be virtually the same.

The Central Limit Theorem^{35,36} (CLT) provides a way to generate a sample that fulfills conditions (i) and (ii). According to this theorem, *the average values, obtained from multiple sets of M independent samples of the same population, will approach a Gaussian distribution. Additionally, the variance of the mean values will be related to the variance of the original samples by a factor $1/M$. Provided that the samples are energy values from a given state of the system, the CLT postulates the possibility of generating a damped Gaussian distribution of energies from each state.* Hence, the exponential average estimator, eq 3, is evaluated using the differences among the average energies, $\langle U \rangle_\nu$, of each state. This protocol is the core of our Central Limit Free Energy Perturbation (CL-FEP) approach.

We implemented this idea in the form of a bootstrapping procedure,³⁷ which operates over the energy values obtained from simulations of the initial and final states of the system, separately. Starting from the original sample of a given state, containing M energy values, every bootstrap cycle selects M values with replacement and computes the corresponding bootstrap average ($\langle U_X \rangle$). Then, the energy difference between the final and initial states is calculated as the difference between the bootstrap means (eq 26). These bootstrap energy differences are normally distributed around the average energy difference between the final and initial states and can be understood as nonequilibrium work values for moving from one state to the other. *Given the general applicability of Jarzynski's equality for nonequilibrium samples, the multiple bootstrap energy differences, ΔU^* , can be used in the evaluation of the exponential average (EXP) estimator without loss of generality.* Thus, the CL-FEP estimator is written as

$$\Delta U_i^* = \langle U_F \rangle_i - \langle U_I \rangle_i \quad (26)$$

$$\Delta \hat{G}_{\text{EXP}} = -kT \ln \langle e^{-\Delta U_i^*/kT} \rangle - B_{\Delta G} + \frac{\sigma_{U_I}^2}{kT} \quad (27)$$

or alternatively, using the second order cumulant (C2) formulation

$$\Delta \hat{G}_{\text{C2}} = \langle \Delta U_i^* \rangle - \frac{\sigma_{\Delta U_i^*}^2}{2kT} + \frac{\sigma_{U_I}^2}{kT} \quad (28)$$

The subscript i represents the i th bootstrap sample. The bias is obtained by using eqs 9 and 11 with the bootstrap energy differences (ΔU^*)

$$B_{(\Delta G)} = \frac{\sigma_{\Delta U_i^*}^2}{2kTN^{\alpha_B}} \quad (29)$$

Thus, the bootstrapping procedure permits a straightforward evaluation of the exponential average estimator based on the CLT. Eqs 26–28 represent the central formalism of the CL-FEP approach. The third term in the right side of eqs 27 and 28 obeys the removal of the thermal heat content of the reference state, $\sigma_{\Delta U^*}^2$, from the variance of the energy differences between initial and final states.

Nevertheless, some issues must be addressed. To begin with, the size (M) of the original energy samples from each state can be arbitrarily large. Thus, in the limit of very large M values the bootstrap averages would erase almost all the variance in the sample. This is to be avoided, as it will remove not only the irrelevant energetic noise but also the variance that is associated with important fluctuations in the host–ligand interactions. Hence, we address the sampling issue within the CL-FEP approach.

To this end, we analyze eq 25 in the context of the bootstrap samples (ΔU^*). Given the fixed value of the free energy variance, the number of bootstrap energy differences, N , would be a function of their corresponding variance, $\sigma_{\Delta U^*}^2$, exclusively

$$N \approx \left(\frac{\sigma_{\Delta U^*}^2}{(kT)^2} \right)^{1/\alpha_V} \quad (30)$$

The variance, $\sigma_{\Delta U^*}^2$, is in turn related to the number of the original energy values from each state, M , and the variance of the original energy samples $\sigma_{\Delta U}^2$

$$\sigma_{\Delta U^*}^2 = \frac{\sigma_{U_F}^2}{M_F} + \frac{\sigma_{U_I}^2}{M_I} \approx \frac{\sigma_{\Delta U}^2}{M} \quad (31)$$

Combining eqs 31 and 25 we obtain

$$(kT)^2 = \frac{\sigma_{\Delta U}^2}{MN^{\alpha_V}} \quad (32)$$

where $\sigma_{\Delta U}^2$ represents the variances of the original energy samples from the initial and final states of the system, the parameter α_V is a function of $\sigma_{\Delta U}^2$ (eqs 11 and 12), M represents the size of the original energy samples from each state, and N is the number of bootstrap iterations (resampled sets).

This way, we consistently address the sampling size problem with eq 32: given that $\sigma_{\Delta U}^2$ and M depend exclusively on the original energy samples, the only parameter that is free to vary is the number of bootstrapping cycles (N). Then, the CL-FEP approach iterates the bootstrapping procedure N times until eq 32 is satisfied. The achievement of this condition is one of the

convergence criteria of the approach. Notably, CL-FEP can converge distinct sets of original energy samples. This means that, instead of the sampling size, it is the relation between the size and the energy variance of sampling what determines the number of iterations required to converge the algorithm. Figure 1 depicts a graphical representation of the CL-FEP approach. In

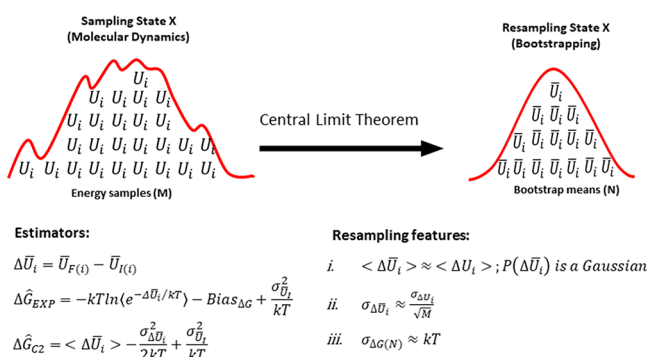


Figure 1. Schematic representation of the Central Limit Free Energy Perturbation (CL-FEP) approach.

the next section we describe the convergence criteria that, in addition to eq 32, are used to determine the final free energy value.

2.2.1. Convergence of the CL-FEP Approach. The CL-FEP algorithm relies on three convergence criteria and one confidence assessment, which depend on the importance sampling ratio obtained during the free energy calculation. These four criteria are automatically evaluated by our code, thus providing information on the convergence of the calculation while estimating the free energy change.

2.2.1.I. Gaussian Sampling. The central limit theorem (CLT) ensures that after a certain number of iterations, the bootstrapping procedure delivers energy differences (ΔU^*) with a normal distribution. Such Gaussian sampling, necessary for the applicability of the formulas describing the free energy bias and variance, is assessed by means of the Shapiro-Wilks normality test.³⁸ This test focuses on the distribution of bootstrap energy differences obtained from the iterations of the bootstrap procedure. The normality of the sample is accepted when the p-value of the Shapiro-Wilks test is larger than 0.05. In the case of failing the test after reaching a maximum number of bootstrap iterations, a “warning type 1” message is issued in the output of the program. This “warning type 1” error should rarely appear, because the CLT ensures the normality of the bootstrap energy differences, typically after the first ten samples (iterations). Failing this test would imply a very small size of the original energy sampling. Thus, this issue can be addressed by increasing the energy samples of each state.

2.2.1.II. The Estimated Standard Deviation (ESD) of the Free Energy Must Be $ESD(\Delta G) \leq kT$. This criterion is related to eq 25, which fixes a constant value for the variance of the free energy estimate. After each step of the bootstrap sampling, the free energy and its expected variance are evaluated. The iterations stop once the normality of the energy samples is achieved and when the ESD reaches the value of kT (i.e., $ESD(\Delta G) \leq kT$). Failing to fulfill this condition within a maximum number of iterations will result in a “warning type 2” message.

Eq 32 shows that reaching this level of variance depends directly on the variance of the original energies, $\sigma_{\Delta U}^2$, and the size

of the original energy sample from each state (M). Thus, for systems with high variance this type of warning could appear if the number of samples M is not sufficiently large. Like for warning type 1, the advised solution is to increase the size of the original samples.

Nevertheless, to improve the feasibility of the approach, we introduced in the algorithm a functionality that performs an oversampling during the resampling step in every iteration of the bootstrap protocol. Within the standard approach for a set of M original energy samples, the code will resample with replacement M energy values to compute an average (eq 26). Using the oversampling functionality, the code will resample $M' = \rho_{OSR} \cdot M$, where ρ_{OSR} represents an oversampling ratio ($\rho_{OSR} \geq 1$). This parameter is by default $\rho_{OSR} = 1$, which renders regular sampling. We evaluated the effectiveness of this oversampling function in the study cases presented below. As discussed, a consequence of the CLT is that the variance of the resampled mean values is related to the variance of the original samples according to eq 31. Given that the oversampling function increases the number of the resampled energy values (M'), eqs 31 and 32 are now written in the form

$$\sigma_{\Delta U^*}^2 = \frac{1}{\rho_{OSR}} \left(\frac{\sigma_{U_F}^2}{M_F} + \frac{\sigma_{U_I}^2}{M_I} \right) \approx \frac{\sigma_{\Delta U}^2}{\rho_{OSR} M} \quad (33)$$

$$(kT)^2 \approx \frac{\sigma_{\Delta U}^2}{\rho_{OSR} M N^{\alpha_v}} \quad (34)$$

The oversampling ratio is an empirical solution to favor the convergence of the variance of the free energy estimate. However, its use is far from being unrestricted as it effectively replaces a missed number of actual energy samples with resampled values. To evaluate the full applicability of the oversampling ratio would require its extensive application to many systems. Nevertheless, our current experiments suggest that ρ_{OSR} values up to 6 provide good free energy estimates.

We note that the oversampling functionality is not an inherent part of the CL-FEP approach. Therefore, we advise caution when large values of ρ_{OSR} are used and suggest increasing the real sampling instead. Currently the code will issue a note when the requested oversampling ratio is $\rho_{OSR} > 6$, alerting that the use of large values is not advised.

2.2.1.III. The Standard Deviation (SD) of the Free Energy Must Be $SD(\Delta G) \leq kT$. The CL-FEP code allows for the execution of several replicas of the CL-FEP procedure (the default amount is 30 replicas). After collecting the final free energy estimate from all replicas, the standard deviation, $SD(\Delta G)$, is obtained and compared with kT that is the level of the expected standard deviation according to eqs 25 and 32. If $SD(\Delta G) > kT$, a “warning type 3” message is launched concerning the lack of convergence. This is the most common warning message that the user may encounter using CL-FEP. Possible solutions depend on how close to convergence is the calculation. An alternative is to increment the maximum number of iterations of the bootstrapping protocol. This way, non-converged independent replicas can attain convergence. Other option is to increase the sampling, which could be achieved by a moderate rise of the oversampling ratio.

2.2.1.IV. Importance Sampling. Conditions I)–III) are hard convergence criteria in CL-FEP. Further, the confidence of the calculated values can be evaluated by means of the importance sampling ratio (ISR). Importance sampling is a key feature for

accurate free energy evaluations¹ and particularly for the exponential average estimator. This average is largely dominated by the lowest values of the energy differences, which are rarely sampled during the calculation. The need to sample the lowest energy differences to properly evaluate the exponential average estimator was early stated by Widom²¹ and subsequently discussed by others authors.^{1,39,40} Hence, the frequency of low energies differences, determined as those values below the final estimated free energy difference, $\Delta U^* < \Delta G$, provides us with a confidence measure for the obtained estimation.

The fraction of samples that CL-FEP evaluates below the final free energy estimate is monitored by the *importance sampling ratio* (ISR). After each run of CL-FEP, the code creates a histogram of the energy differences sampled during the run and prints the calculated ISR. Larger ISR values correspond to more reliable free energy estimates. The current implementation of CL-FEP prints an alert note when the ISR is very low ($\text{ISR} < 2.5\%$). In such cases, converged estimates with larger ISR should be reached by either augmenting the original sampling or the oversampling ratio.

In the CL-FEP analysis, the input data is energy samples from the initial and final states of the system. The CL-FEP code is designed to split the input samples into multiple subsets of increasing size, which are identified as checkpoints for convergence. The first checkpoint is always placed above 25% of the input energy samples of each state. The exact amount of sampling in each checkpoint is $(25 + 75n/m)\%$, where m is the total number of checkpoints, and n is the index of that specific checkpoint. Here, we run CL-FEP requesting 15 checkpoints ($m = 15$), which means that the first checkpoint ($n = 1$) contains 30% of the sampling of each state while the tenth checkpoint encompasses 75% of the sampling of each state. For all checkpoints, the free energy is estimated together with the convergence and confidence analysis. We advise to always set a high number of checkpoints (≥ 10) and to extend the sampling, or the oversampling ratio, until convergence of at least one-third of the checkpoints. This allows verifying that the converged estimations of the free energy change do not vary, significantly, with the increase of the sampling.

A combined estimator is also computed by the code, in the form of a weighted-average free energy change

$$\Delta G^{\text{ave}} = \frac{\sum \text{ISR}_i \cdot \Delta G_i}{\sum \text{ISR}_i} \quad (35)$$

where the sums extend along all the *converged* checkpoints.

2.2.2. CL-FEP Implementation. CLFEP.pl is a postprocessing script, written in Perl, which receives as input the direct log files produced by simulations with the NAMD 2.12 (or 2.13) code.^{41,42} In the case of using a different simulation package, a comma separated values (CSV) file can be utilized by listing, in each line, three columns with (i) the potential energy value, (ii) the contribution from external restraints to the specific energy value, and (iii) the box volume in the corresponding frame of the simulation. If no restraints are used, or the system stays within wall potentials where no penalization is applied, the second column must have a zero in each line. In this implementation, we subtract the restraints contribution (if any) from the corresponding total potential energy values, before the evaluation of the free energy estimators. In [Supplementary File 1](#) we provide the input files used to run the CL-FEP analysis for the study cases presented here, as well as the corresponding output files.

In [Scheme 1](#), we present a pseudocode describing the main steps of the CL-FEP algorithm.

Scheme 1. Pseudocode of the CLFEP.pl Implementation

```

Foreach state (initial, final) do:
  Read energy and volume values from input files;
  Randomize the energy samples;
End Foreach;
Compute the finite-size effect correction; # Supp. Mat.
Compute the standard-state correction; # Supp. Mat.
Foreach checkpoint do:
  Foreach state (initial, final) do:
    Extract subsample of energies values based on current checkpoint;
  End Foreach;
  Create total energy list;
  Foreach replica do:
    Create energy list;
    While ( $\sigma(\Delta G) > kT$  and (energy list NOT normal)): # Start Bootstrapping
      Foreach state (initial, final) do:
        Perform a random resampling with replacement;
        Compute average energy ( $\langle U \rangle$ );
      End Foreach;
      Compute the resampled energy difference ( $\Delta U^*$ ); # Eq. 26
      Apply finite-size effect correction to  $\Delta U^*$ ;
      Add current  $\Delta U^*$  value to the energy list;
      Obtain the p-value of Shapiro-Wilks test on energy list;
      Compute free energy difference ( $\Delta G$ ) using the energy list; # Eqs. 27 and 28
      Compute the estimated standard deviation  $\sigma(\Delta G)$ ; # Eq. 10
      Compute the estimated error RMSE( $\Delta G$ ); # Eq. 15
      Save to output  $\Delta G$ ,  $\sigma(\Delta G)$ , RMSE( $\Delta G$ ),  $\Delta U^*$ , p-value;
    End While; # End Bootstrapping
    Add energy list to the total energy list;
  End Foreach;
  Compute average values from the final output of every replica;
  Compute the standard deviation SD( $\Delta G$ ) among replicas;
  Compute the importance sampling ratio ISR using the total energy list;
  Save to final output  $\langle \Delta G \rangle$ ,  $\langle \sigma(\Delta G) \rangle$ , SD( $\Delta G$ ),  $\langle \text{RMSE}(\Delta G) \rangle$ ,  $\Delta H$ ,  $\sigma(\Delta H)$ ,  $\langle p\text{-value} \rangle$ , ISR;
Do convergence analysis:
   $\langle p\text{-value}(\Delta U^*) \rangle > 0.05$  ? else Warning type 1;
   $\langle \sigma(\Delta G) \rangle < kT$  ? else Warning type 2;
  SD( $\Delta G$ ) < kT ? else Warning type 3;
End Do;
End Foreach;
Compute the weighted free energy average among converged checkpoints; # Eq. 35
End Program;

```

CLFEP.pl is implemented with shared memory parallelization, which allows runtimes of a few minutes in multicore workstations and is available upon request. The CL-FEP analysis will be accessible as a web server implementation at <https://clfep.zmb.uni-due.de/>.

3. APPLICATIONS OF CL-FEP TO COMPUTE BINDING FREE ENERGIES

3.1. Molecular Dynamics Sampling. We first studied four benchmark systems ([Figure 2](#)), whose binding free energies were previously calculated using the CHARMM force field^{43,44} in good agreement with experimental values.^{45–52}

We used the initial coordinates and parameters of the cyclodextrin–progesterone complex, the SH3–Ac-APSYSPPP-PP system, and the 18-crown-6 (18C6) ether bound to a potassium cation (K^+) previously employed by Chipot et al.^{45,48,53} The coordinates for complex SH2–Ac-pYEEI were taken from the study of Woo and Roux.^{52,54} The sampling of the initial and final states was performed with independent molecular dynamics simulations (with same temperature and pressure control) on subsystems containing the ligand, the host, the complex, and the bulk solvent. As done previously by Chipot et al.,³⁶ the potassium cation in the 18C6– K^+ complex was restrained with respect to the center of mass of the 18-crown-6 ether, using a wall potential, in order to prevent it from escaping the host during the simulations. In all cases, the simulations were performed with NAMD.^{41,42} The PME method⁵⁵ was employed to account for long-range electrostatic interactions. The van der Waals interactions were truncated at 12 Å without the switching

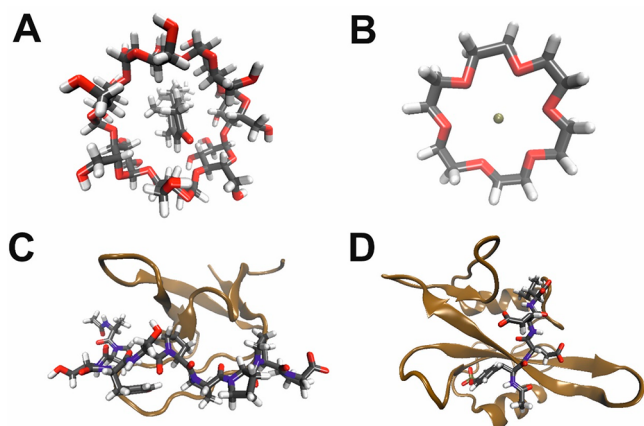


Figure 2. Benchmark systems used to assess the accuracy of the CL-FEP approach. A) Complex between cyclodextrin and progesterone. B) Complex between the 18-crown-6 ether and a potassium cation. C) Complex between the SH3 domain of Abl kinase (cartoon) and the peptide Ac-APSYSPPPPP (licorice). D) Complex between the SH2 domain of Abl kinase (cartoon) and the peptide Ac-pYEEI (licorice).

procedure. The systems were solvated using the TIP3P water model,⁵⁶ and the same number of water molecules was used in each of the boxes representing the individual states (complex, guest, host and solvent). Ions were added for neutralization. The bulk solvent simulations were set with the amount of ion pairs necessary to accomplish the balance of counterions between reactants and products. The simulations were equilibrated in two steps: first we performed an NVT simulation for 100 ps allowing only water molecules and ions to move, followed by 5 ns of NPT simulation with harmonic constraints on the heavy atoms of the solute. Production runs (NPT) were carried out using Langevin dynamics⁵⁷ with a damping constant of 5 ps^{−1}. The energy sampling of the initial and final states of each system was carried out by means of several simulations (up to 50 ns) starting from the same geometry but using different initial atomic velocities. The potential energy was monitored every 5 ps.

For each subsystem, M energy samples are collected. The CL-FEP protocol is then executed by bootstrapping every subset multiple times and obtaining the bootstrap average corresponding to each iteration. Subsequently, the bootstrap energy differences to evaluate the free energy estimator are calculated as

$$\Delta U_{bind_i}^* = \langle U_C \rangle_i + \langle U_S \rangle_i - \langle U_H \rangle_i - \langle U_L \rangle_i \quad (36)$$

where the subscript i represents the i th bootstrap iteration. U_C stands for the potential energy sample in the box with the complex. U_S corresponds to the sample in the bulk solvent box, U_H corresponds to the sample of the host, and U_L represents the sample of the ligand (Figure 3). Then, the exponential average or the second order cumulant estimators (eqs 27 and 28) are used to obtain the free energy estimate. The CL-FEP analysis was performed with 15 checkpoints. In each checkpoint the CL-FEP procedure was replicated 50 times using a maximum of 2000 bootstrapping iterations. The final estimate of the free energy change was determined based on the average among converged checkpoints (eq 35).

Remarkably, CL-FEP reproduced the binding affinities of all systems with deviations lower than 1 kcal/mol (Table 1). This accuracy matches that of more demanding free energy approaches, which exploit either geometrical or alchemical pathways to evaluate free energy changes.^{45,46,48,51}

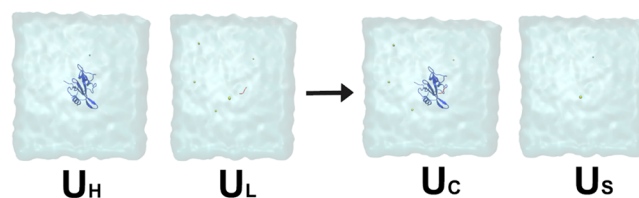


Figure 3. Decomposition of the sampling process into four equilibrium subsystems to compute binding free energies. U_H corresponds to the sample of the host, and U_L corresponds to the sample of the ligand. U_C corresponds to the potential energy sample in the box with the complex, and U_S corresponds to the sample in the bulk solvent box.

Table 1. Summary of Experimental and Calculated Binding Free Energies Using CL-FEP with the Exponential Average (EXP) and the Second Order Cumulant (C2) Estimators

system	experimental	calculated	
	ΔG^0 (kcal/mol)	ΔG_{EXP}^{ave} (kcal/mol)	ΔG_{C2}^{ave} (kcal/mol)
CD-PRO	−5.6 ⁵⁸	−5.1 ± 0.54	−5.2 ± 0.51
18C6-K ⁺	−2.91 ⁵⁹	−3.9 ± 0.45	−3.4 ± 0.46
Abl_SH3-Ac-APSYSPPPPP	−7.99 ^{60,61}	−8.2 ± 0.51	−7.5 ± 0.53
Abl_SH2-Ac-pYEEI	−7.1 ⁶²	−6.5 ± 0.58	−7.0 ± 0.54

3.2. Complex of Cyclodextrin (CD) with Progesterone (PRO). The experimental binding free energy of the cyclodextrin–progesterone (CD-PRO) complex is reported as −5.6 kcal/mol.⁵⁸ Chipot et al. used this system involving two rigid neutral molecules to introduce BFEE,⁴⁵ a graphical user interface associated with a multistep methodology based on eABF calculations. Their calculated binding free energy was −5.3 kcal/mol,⁴⁵ which improved a previous result (−6.6 kcal/mol⁴⁶) on the same system.

Despite its small size and the rigid nature of its components, the CD-PRO complex (Figure 2A) displays a considerable energetic variance. This is because the progesterone molecule can adopt multiple poses within a varying degree of penetration in the host, thus spreading the range of binding energies. To obtain a high number of converged checkpoints, the over-sampling ratio was fixed at $\rho_{OSR} = 2$. In Table 2 we list the free energy estimates from the 15 checkpoints, along with information on convergence and importance sampling ratio. These values were obtained using the exponential average estimator, which is the native estimator in FEP¹¹ (see Supporting Information, Table S1 for the results with the second order cumulant variant). Eight out of the 15 checkpoints did not converge, given that the standard deviations of the free energy estimates among the 50 replicas of the CL-FEP were larger than 0.593 kcal/mol (>RT). This corresponds to a warning message type 3 (WARNING3). Notably, the estimation based on the weighted average of the checkpoints (−5.1 kcal/mol) deviates from the experimental value by less than the RT value.

Similarly, the second order cumulant estimator rendered nine converged checkpoints (Supporting Information, Table S1). The prediction based on the second order cumulant was −5.2 kcal/mol, in good agreement with the exponential average value. We also note that the predictions with both estimators (−5.1 and −5.2 kcal/mol) are very similar to the result reported using the state-of-the-art eABF approach (−5.3 kcal/mol).⁴⁵

3.3. 18-Crown-6 Ether (18C6) Bound to a Potassium Cation. The experimental binding affinity of this complex is

Table 2. Thermodynamic and Convergence Parameters Obtained for All the Checkpoints in the System CD-PRO Using the Exponential Average Estimator (EXP)

point (%)	ΔG_{EXP} (kcal/mol)	ISR (%)	convergence
1 (30)	-8.4 ± 0.79	11.11	WARNING3
2 (35)	-6.6 ± 0.79	18.87	WARNING3
3 (40)	-6.5 ± 0.66	16.17	WARNING3
4 (45)	-4.3 ± 0.80	23.52	WARNING3
5 (50)	-5.1 ± 0.64	23.31	OK
6 (55)	-5.3 ± 0.60	24.62	OK
7 (60)	-5.3 ± 0.70	25.2	WARNING3
8 (65)	-5.8 ± 0.62	26.32	WARNING3
9 (70)	-5.7 ± 0.68	25.69	WARNING3
10 (75)	-5.4 ± 0.50	25.58	OK
11 (80)	-5.4 ± 0.62	18.58	WARNING3
12 (85)	-5.2 ± 0.58	21.01	OK
13 (90)	-5.1 ± 0.45	22.29	OK
14 (95)	-5.1 ± 0.55	28.33 ^a	OK
15 (100)	-4.6 ± 0.48	25.93	OK
average	-5.1 ± 0.54		

^aLargest value of the importance sampling ratio (ISR) among the converged checkpoints.

reported as -2.91 kcal/mol.⁵⁹ Chipot et al. used this system first to introduce ParseFEP, a toolkit to process FEP calculations, and reported a free energy estimate of -4.0 ± 0.1 kcal/mol.⁴⁷ Later, they used again the 18C6-K⁺ complex for an application of stratified FEP calculations within the NAMD program. There, their estimate was -2.6 kcal/mol.⁴⁸ Such a value also represented an improvement with respect to early calculations reported by Dang and Kollman (-3.5 kcal/mol⁴⁹ and -2.0 kcal/mol⁵⁰) based on potential of mean force calculations from molecular dynamics simulations.

Like in the previous case, the host molecule does not show high conformational variability (Figure 2B). However, even with small changes in the geometry of the complex, the charged guest cation imposes large fluctuations to the energy. In addition, the interactions of potassium with its counterion are a source of energy fluctuations. Hence, the oversampling ratio was fixed at $\rho_{\text{OSR}} = 2$ in order to obtain several converged checkpoints.

Four out of the 15 checkpoints did not achieve convergence, with a warning message type 3 (Table 3). As mentioned above, this is the most frequent reason for nonconverged checkpoints and is usually solved by increasing the sampling, which is evidenced by the convergence of the later checkpoints. Interestingly, our case studies show a common trend illustrating the better convergence properties of the second order cumulant estimator over the exponential average (Supporting Information, Tables S1–S4 and Tables 2–5, respectively).

In the present study, the exponential average overestimated the binding affinity of 18C6-K⁺ by nearly 1.0 kcal/mol, rendering a prediction of -3.9 kcal/mol based on the weighted average of the checkpoints. The free energy change computed with the second order cumulant (-3.4 kcal/mol) is in closer agreement to the experimental value (Supporting Information, Table S2). The linear average in the second order cumulant estimator (eq 28) is less affected by extreme values in the distributions than the exponential average (eq 27). Consequently, our result suggests that there is some noisy energetic variance (like that from the bulk solvent) that is still not completely damped by the CL-FEP approach. This could be because the energy fluctuations arising from the interactions

Table 3. Thermodynamic and Convergence Parameters Obtained for All the Checkpoints in the 18C6-K⁺ System Using the Exponential Average Estimator (EXP)

point (%)	ΔG_{EXP} (kcal/mol)	ISR (%)	convergence
1 (30)	-5.9 ± 0.66	31.58	WARNING3
2 (35)	-5.1 ± 0.78	14.03	WARNING3
3 (40)	-4.2 ± 0.63	47.94	WARNING3
4 (45)	-3.8 ± 0.59	29.98	WARNING3
5 (50)	-4.2 ± 0.52	21.22	OK
6 (55)	-4.3 ± 0.49	17.96	OK
7 (60)	-3.7 ± 0.49	22.36	OK
8 (65)	-3.0 ± 0.43	31.17	OK
9 (70)	-3.8 ± 0.50	27.62	OK
10 (75)	-4.0 ± 0.48	16.39	OK
11 (80)	-4.4 ± 0.47	26.81	OK
12 (85)	-4.2 ± 0.45	58.09 ^a	OK
13 (90)	-4.3 ± 0.39	27.13	OK
14 (95)	-3.8 ± 0.39	19.04	OK
15 (100)	-3.8 ± 0.38	29.20	OK
average	-3.9 ± 0.45		

^aLargest value of the importance sampling ratio among the converged checkpoints.

between the potassium cation and its counterion are larger than the most relevant interactions between the cation and the crown ether. Hence, the result of the damping is still affected by some of that irrelevant variance, leading to a better performance of the second order cumulant estimator.

Importantly, although charged species may produce higher noise in the energy samples, CL-FEP is still capable of extracting the relevant variance out of that noise and use it for an accurate estimation of the free energy difference. Particularly, the second order cumulant estimator seems more stable for systems with large energetic noise. This result, together with the previous case study, evidences the suitability of CL-FEP to estimate the binding affinity of host-guest complexes with both molecular and ionic ligands.

3.4. Src Homology 3 Domain (SH3) of Abl Kinase Bound to Ac-APSYSPPPPP. The experimental binding affinity of the SH3-Ac-APSYSPPPPP complex is reported as -7.99 kcal/mol.^{60,61} This system was also used by Chipot et al. to introduce the BFEE tool.⁴⁵ In addition, in a previous study by the same group, a stratified FEP calculation was used to compare the accuracy of both a geometrical (eABF) and an alchemical route to compute free energy changes.⁵¹ Using both the geometrical and the alchemical routes, the authors succeeded in reproducing the absolute binding free energy, reporting values of -7.8 and -7.7 kcal/mol, respectively.⁵¹ Here we show that CL-FEP delivers a similar level of accuracy.

SH3-Ac-APSYSPPPPP (Figure 2C) is a complex between a small protein domain (SH3 domain of Abl kinase) and a 10-residue proline-rich peptide. The flexibility of both the peptide and the protein loops makes this system a good model to challenge the performance of CL-FEP in protein-peptide interactions.

Due to their relevance in the context of our approach, we briefly outline specific details of the simulations of this system. The dynamics simulations were prepared using the same number of water molecules in all boxes, and ions were initially added for neutralization. As a result, one Na⁺ is added to the peptide box, another Na⁺ is placed in the protein box, and two Na⁺ are in the box of the complex. We note an important aspect

regarding the counterions in the setup of simulation boxes for CL-FEP: the so far obtained distribution of charged species, even when it renders neutral simulation boxes, produces an artificial excess of chemical potential. This is because, in the initial states, four charged species (protein, ligand, and two counterions) are distributed evenly in two boxes. The situation is different for the products, where all the charged species are confined in a single box while the box with the bulk solvent lacks ions. Thus, we note that simply neutralizing each of the simulation boxes is insufficient to achieve charge balance. To prevent a bias in the chemical potential of the charged species, we created an even distribution of the number of charged species in the initial state and final state boxes. To this end, we added a sodium chloride (NaCl) molecule to each of the boxes in the initial states and two NaCl molecules to the product box with the bulk solvent. Thus, we had four charged species in each simulation box.

As mentioned above, the SH3-Ac-APSYPPPPP system adds an additional level of complexity to our performance assessment: we first started with a neutral and rigid complex (CD-PRO), then moved into a charged but seemingly rigid system (18C6-K⁺), and now investigate a system featuring conformational flexibility in addition to charged species. Hence, the oversampling ratio was fixed at $\rho_{\text{OSR}} = 3$. Using the exponential average estimator, we obtained six converged checkpoints, which delivered a free energy change of -8.2 kcal/mol (Table 4), in excellent agreement with the experimental binding free energy (-7.99 kcal/mol).

Table 4. Thermodynamic and Convergence Parameters Obtained for All the Checkpoints in the System SH3-Ac-APSYPPPPP Using the Exponential Average Estimator (EXP)

point (%)	ΔG_{EXP} (kcal/mol)	ISR (%)	convergence
1 (30)	-7.0 ± 0.80	12.72	WARNING3
2 (35)	-5.7 ± 0.77	15.75	WARNING3
3 (40)	-6.1 ± 0.81	23.01	WARNING3
4 (45)	-6.0 ± 0.70	30.50	WARNING3
5 (50)	-5.9 ± 0.77	26.38	WARNING3
6 (55)	-7.3 ± 0.75	26.68	WARNING3
7 (60)	-7.2 ± 0.62	25.78	WARNING3
8 (65)	-6.1 ± 0.69	35.23	WARNING3
9 (70)	-7.6 ± 0.60	27.21	WARNING3
10 (75)	-7.5 ± 0.56	29.71	OK
11 (80)	-7.8 ± 0.53	20.63	OK
12 (85)	-8.3 ± 0.50	33.06	OK
13 (90)	-8.7 ± 0.49	18.09	OK
14 (95)	-8.7 ± 0.52	33.74 ^a	OK
15 (100)	-8.4 ± 0.46	21.86	OK
average	-8.2 ± 0.51		

^aLargest value of the importance sampling ratio among the converged checkpoints.

With the second order cumulant estimator, CL-FEP rendered nine converged checkpoints with average binding free energy of -7.5 kcal/mol (Supporting Information, Table S3). This estimation is also very close to the experimental measure, thus supporting the applicability of CL-FEP to compute binding free energies of protein–ligand complexes with accuracy. Like with the small host–guest complexes, the C2 estimator shows better convergence properties.

3.5. Src Homology 2 Domain (SH2) of Abl Kinase Bound to Ac-pYEEI. The experimental binding free energy of the SH2-Ac-pYEEI complex is reported as -7.1 kcal/mol.⁶² This system was used by Woo and Roux to illustrate a multistep approach to compute absolute binding free energies combining stratified FEP and umbrella sampling simulations.⁵² They obtained a binding free energy of -8.8 kcal/mol, in close agreement with the experimental value. By contrast, the same authors reported an MMPBSA-based estimation of -82 ± 20 kcal/mol for the binding free energy of the this system.⁵²

The SH2-Ac-pYEEI complex (Figure 2D) displays a higher level of complexity with respect to our previously discussed test cases. Here, a highly charged phosphorylated tetrapeptide (five negative charges) is interacting with an also charged protein domain (one positive charge). Both the flexibility of the peptide and its charge produce a large energetic noise. To surpass this challenge within the few hundred nanoseconds sampling scale, the oversampling ratio was increased up to $\rho_{\text{OSR}} = 6$ to converge multiple checkpoints.

Like in the previous case, we briefly discuss some details of the simulation's setup relevant to the CL-FEP outcome. The simulations boxes were prepared with the same number of water molecules, and ions were initially added for neutralization. However, the mass balance among the boxes is not complete after neutralizing each system, since the initial state boxes require 5 Na⁺ (ligand) and 1 Cl[−] (protein), while the complex only needs 4 Na⁺. Hence, an additional NaCl molecule was placed in the solvent box, to avoid an artificial excess chemical potential. Thus, we have 10 charges in the ligand box, which are paired with 10 charges in the complex box, as well as 2 charges in the protein box and 2 charges added in the solvent box. The charges are counted based on the net charge of each specie, i.e., the protein is counted as a single entity with a charge equal to the net charge considering all its residues.

After running the CL-FEP analysis, we obtained nine converged checkpoints using the exponential average estimator (Table 5). The weighted average based on the converged checkpoints results in a free energy change estimation of -6.5 kcal/mol.

Using the second order cumulant estimator, we obtained nine converged checkpoints too, whose weighted average rendered -7.0 kcal/mol, in excellent agreement with the experimental value of -7.1 kcal/mol⁶² (Supporting Information, Table S4). Thus, CL-FEP is also capable of reproducing the experimental binding affinity of protein–peptide complexes even when highly charged and flexible species are involved.

The case studies presented in this work were selected to challenge the applicability of CL-FEP in different scenarios of varying complexity. Remarkably, the free energy evaluations showed deviations that are within 1 kcal/mol from the experimental values and very close to the estimated error of the method. In general, the second order cumulant estimator converges easier than the exponential average, and it shows slightly better accuracy, although the relative improvement is within the error of the estimations.

3.6. Performance of CL-FEP in Benchmark Sets of Complexes. To further validate our methodology, we employed additional benchmark sets:

- Seven binders to the T4 lysozyme (lysozyme L99A/M102Q set) were taken from the review of Mobley and Gilson.⁶³

Table 5. Thermodynamic and Convergence Parameters Obtained for All the Checkpoints in the SH2–Ac- pYEEI System Using the Exponential Average Estimator (EXP)

point (%)	ΔG_{EXP} (kcal/mol)	ISR (%)	convergence
1 (30)	-3.8 ± 0.85	17.25	WARNING3
2 (35)	-2.9 ± 0.83	17.24	WARNING3
3 (40)	-2.8 ± 0.82	24.8	WARNING3
4 (45)	-3.0 ± 0.76	18.99	WARNING3
5 (50)	-5.2 ± 0.63	22.31	OK
6 (55)	-5.2 ± 0.67	25.27	WARNING3
7 (60)	-4.9 ± 0.62	14.99	OK
8 (65)	-6.2 ± 0.65	29.35	WARNING3
9 (70)	-7.3 ± 0.58	20.97	OK
10 (75)	-7.0 ± 0.59	22.81	OK
11 (80)	-6.5 ± 0.58	14.67	OK
12 (85)	-7.1 ± 0.55	41.15 ^a	OK
13 (90)	-7.0 ± 0.56	25.39	OK
14 (95)	-6.6 ± 0.56	26.52	OK
15 (100)	-5.9 ± 0.59	31.25	OK
average	-6.5 ± 0.58		

^aLargest value of the importance sampling ratio among the converged checkpoints.

- ii) Eight binders to octa-acid (OA) and tetra-methyl-endo octa-acid (TEMOA) targets, as used in the SAMPL6 challenge of binding affinity prediction.⁶⁴
- iii) Two inclusion complexes between the molecular tweezer CLR01 and the amino acids lysine and arginine (Figure 4). CLR01 is a supramolecular organic compound that selectively binds lysine and, to less extent, arginine residues via a combination of hydrophobic and electrostatic interactions.⁶⁵ As host molecule, CLR01 has interesting features, involving two hydrogen phosphate groups that can introduce a pH dependency in the binding of small ligands, also a semirigid hydrophobic cavity able

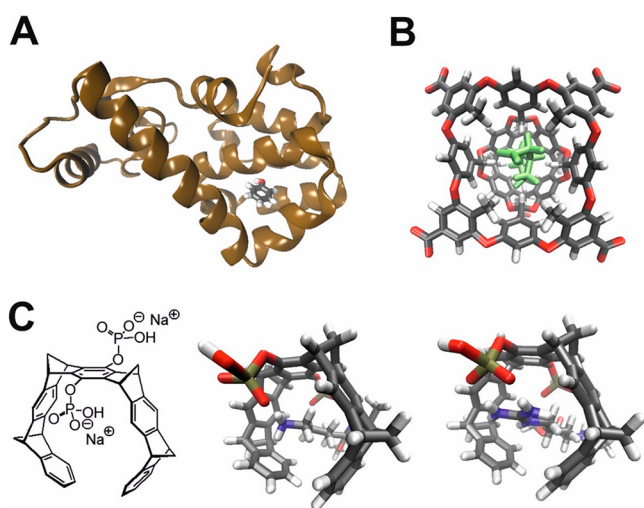


Figure 4. Representative structures corresponding to the benchmark sets (i)–(iii). A) T4 lysozyme (cartoon) with phenol (licorice). B) Tetra-methyl-endo octa-acid (TEMOA, licorice) with cyclopentyl-acetic acid (green licorice). C) Structure of the molecular tweezer CLR01 (left) and its complexes with lysine (middle) and arginine (right). The side chain of the amino acid was inserted in the cavity of the tweezer forming a salt bridge with one of the hydrogen phosphate groups, as reported in previous studies.^{67–70}

to accommodate ligands of slightly different size and shape. The guest molecule in the cavity of CLR01 is partially desolvated, which indicates that most of the affinity comes from a fine balance of hydrophobic and ionic host–guest interactions. The binding affinity of phosphorus-containing molecular tweezers, measured using ITC experiments in 200 mM NaH_2PO_4 (pH = 7.6), is highly dependent on the ionic strength in the experiments.^{65,66} In addition, CLR01 is reported as inhibitor of pathogenic aggregation and as regulator of protein–protein interactions.^{67–70} Thus, CLR01–Arg/Lys complexes offer an interesting and challenging platform to assess new binding free energy methodologies.

Combined, sets (i)–(iii) provide 25 additional complexes, which complements the applicability studies of CL-FEP discussed in the previous section.

All the systems were sampled following the molecular dynamics protocol described in Section 3.1. Sets (ii) and (iii) and the ligands of set (i) were modeled using the CHARMM General Force Field parameters⁷¹ and the CGenFF web server.^{72,73} For adding solvent and ions, we implemented a new tool, CLFEP_autoconf.tcl, which is a VMD⁷⁴ script that can be obtained along with our CLFEP.pl code. CLFEP_autoconf.tcl aims to guarantee the construction of cubic simulation boxes with an even distribution of water molecules (among all the boxes) as well as a proper balance of counterions and a fixed total charge concentration. This is achieved by adjusting both the number of ions and the volume of the systems. The ionic concentrations were 20 mM and 200 mM for sets (ii) and (iii), respectively. Set (i) was modeled at 500 mM in analogy to the NaCl concentration used in ITC measurements.⁷⁵ The structure with PDB code 1LI3⁷⁵ was used to build all lysozyme complexes. A harmonic wall potential was applied to the distance between the center of masses of the host and the ligand in the bound states of sets (ii) and (iii); the boundary was placed 4 Å away of that distance in the complex structures.

For all systems, we ran 10 independent analyses of CL-FEP using the C2 estimator, which delivers the best convergence, as discussed above. Each run consisted of 10 checkpoints with 30 internal replicas. The ρ_{OSR} value was fixed to obtain at least half of the checkpoints converged (see footnotes in Table 6). In each run the results of the converged checkpoints were combined, using eq 35, to obtain the corresponding estimate. Finally, averages and standard deviations were calculated. The low values (<1 kcal/mol) of the standard deviations among the results of independent runs of the CL-FEP analysis evidence the convergence of the setup.

Set (i), T4 Lysozyme L99A/M102Q. Notably, the predicted free energies of binding of most of the compounds in this set are very close to the experimental values (Table 6). We found the larger deviations for phenol, catechol, and 3-chlorophenol, the smallest ligands. This can be rationalized by the fact that these compounds have larger freedom in the binding site (Figure 4A), which causes an increased variance of the energy in the bound state. The second term in eq 28 evidences that a high variance favors more negative estimations of the binding energy (e.g., high entropy contribution to the binding). Importantly, the mean absolute error (MAE) for this set of complexes is 1.1 kcal/mol, which is within the previously reported range of 1–2 kcal/mol, as reviewed by Mobley and Gilson.⁶³

Table 6. Summary of Calculated and Experimental Free Energies of Binding for the Ligands of T4 Lysozyme L99A/M102Q, the Octa Acids OA and TMOA, and the Molecular Tweezer CLR01^a

ligand	ΔG_{calc}	$\Delta G^{**}_{\text{EXP}}$	abs error
T4 lysozyme^a			
phenol	-8.0 ± 0.62	$-5.24 \pm \text{NR}$	2.8
catechol	-6.7 ± 0.42	-4.16 ± 0.03	2.5
2-ethoxyphenol	-3.6 ± 0.81	-4.02 ± 0.03	0.4
benzyl acetate	-4.1 ± 0.38	-4.48 ± 0.16	0.4
4,5,6,7-tetrahydroindole	-4.5 ± 0.44	-4.61 ± 0.09	0.1
N-phenylglycinonitrile	-5.8 ± 0.69	-5.52 ± 0.18	0.3
3-chlorophenol	-4.1 ± 0.81	$-5.51 \pm \text{NR}$	1.4
mean			1.1
OA^b			
G0: cyclopentyl acetic acid	-4.8 ± 0.46	-5.68	0.9
G1: <i>trans</i> -2-hexenoic acid	-2.7 ± 0.61	-4.56	1.9
G2: (S)-perillic acid	-6.3 ± 0.61	-8.38	2.1
G3: 5-hexenoic acid	-3.8 ± 0.66	-5.18	1.4
G4: (S)-citronellic acid	-6.5 ± 0.51	-7.11	0.6
G5: 2-methyl-4-pentenoic acid	-5.1 ± 0.59	-4.59	0.5
G6: 4-methylpentanoic acid	-5.1 ± 0.69	-4.97	0.1
G7: 2,2-dimethyl-4-pentenoic acid	-8.3 ± 0.53	-6.22	2.1
mean			1.2
TEMOA^b			
G0: cyclopentyl acetic acid	-7.0 ± 0.76	-6.06	0.9
G1: <i>trans</i> -2-hexenoic acid	-3.9 ± 0.71	-5.97	2.1
G2: (S)-perillic acid	-5.2 ± 0.72	-6.81	1.6
G3: 5-hexenoic acid	-6.7 ± 0.65	-5.60	1.1
G4: (S)-citronellic acid	-7.9 ± 0.50	-7.79	0.1
G5: 2-methyl-4-pentenoic acid	-3.1 ± 0.71	-4.16	1.1
G6: 4-methylpentanoic acid	-5.2 ± 0.53	-5.40	0.2
G7: 2,2-dimethyl-4-pentenoic acid	-4.4 ± 0.82	-4.13	0.3
mean			1.1
CLR01^c			
Ac-Lys-OMe	-5.5 ± 0.40	-6.5 ± 0.2	1.0
Ac-Arg-OMe	-4.0 ± 0.63	-5.76 ± 0.02	1.8
mean			1.4

^aThe sampling was performed using 200 ns simulation time for each subsystem (complex, host, ligand, solvent). In all cases, the CL-FEP analysis was run 10 times using $\rho_{\text{OSR}} = 2$, and the reported values correspond to the mean and standard deviation among these runs. Experimental values were taken from Mobley and Gilson.⁶³ NR: not reported. ^bThe sampling was performed using 150 ns simulation time for each subsystem (complex, host, ligand, solvent). In all cases, the CL-FEP analysis was run 10 times using $\rho_{\text{OSR}} = 5$, and the reported values correspond to the mean and standard deviation among the runs. Experimental values were taken from Rizzi et al.⁶⁴ ^cThe sampling was performed using 100 ns simulation time for each subsystem (complex, host, ligand, solvent). In all cases, the CL-FEP analysis was run 10 times using $\rho_{\text{OSR}} = 3$, and the reported values correspond to the mean and standard deviation among the runs. Experimental values were taken from Dutt et al.⁶⁵ ^dAll values are in kcal/mol.

Set (ii), Octa Acids OA and TEMOA. The MAE of the calculated affinity of the ligands for the octa acids (OA and TEMOA) is 1.1 kcal/mol (Table 6), which is in very good agreement with the MAE values reported using FEP_MM and FEP_QM/MM calculations (SAMPL6 challenge: 1.3 and 1.4 kcal/mol, respectively).⁶⁴ Interestingly, calculations presented

in the SAMPL6 event with the end-state MMPBSA approach rendered an MAE of 6.4 kcal/mol for the same benchmark set.^{64,76}

Of both octa acids, OA shows the larger deviations, which may be related to limitations in the sampling due to its larger pocket, which lacks the four endomethyl groups that narrow the entry of the pocket (Figure 4B). The binding of these systems is driven by weak van der Waals interactions that together with the wide and open binding pocket make necessary the use of a boundary potential to maintain the sampling within bound state geometries during the simulations of the complexes. In addition, the re-evaluation of the energies without the switching function in the Lennard-Jones potential allowed recovering much of the binding free energy, evidencing the importance of dispersive interactions for the stability of these complexes.

Set (iii), Molecular Tweezer CLR01. The complexes between the molecular tweezer CLR01 and the amino acids lysine and arginine are challenging systems (Figure 4C). In both residues, the N-terminal amino group was acetylated, and the charge at the C-terminus was neutralized via methylation. As in other studies,^{67–70} CLR01 was modeled with two hydrogen phosphate groups as substituents and thus a net charge of -2 .

Remarkably, CL-FEP achieves a very good evaluation of the binding affinity of both complexes. Particularly for lysine, the accuracy is enhanced by preventing the use of switching functions for van der Waals interactions. Consequently, the Lennard-Jones potential is fully described within the cutoff distance, thus properly representing the contribution of the dispersive interactions in the system. This fact indicates that, in agreement with experimental observations,⁶⁵ the hydrophobic interactions between the tweezer's cavity and the four methylene groups of the lysine's side chain play a key role in the stability of the complex. The larger deviation observed for the arginine complex can be related to shortcomings in the force field parameters to model the tweezer's scaffold, as the guanidinium group enforces a wider tweezer's cavity.

Taken together, these results indicate that our approach, developed to solve the exponential average estimator (FEP/NEW equation) from the end-state energies of a system transformation, delivers the same level of accuracy of traditional stratified solutions of the FEP identity or the umbrella sampling estimator.⁶⁴ Nonetheless, some practical aspects must be taken into account for the application of the CL-FEP protocol.

3.7. Practical Remarks. Here, we discuss practical details relevant for the application of the method.

3.7.I. Mass Balance. The number of water molecules and counterions must be the same between the initial and final states in order to avoid an excess chemical potential due to the change in the composition of the system. As for the water molecules, an effective way to achieve balance is to keep a pairwise correspondence in the number of water molecules between the larger boxes, i.e., the protein and the complex simulation box, and the smaller ones, i.e., the ligand and the bulk solvent box.

3.7.II. Detailed Charge Balance. For several systems, such as the T4 lysozyme, the predictions of absolute binding free energies are very much affected by the total ionic concentration. Thus, this feature should be properly fixed when building the simulation boxes, by resembling as much as possible the experimental conditions. An important guideline for the preparation of the systems is that the number of charges should be the same among the simulation boxes. This condition is achieved by introducing counterions, in addition to those needed for neutralization, in a way that all the simulation boxes

possess the same numbers of net positive and negative charges. Additionally, the volume should be fixed in a way that the total charge concentrations adopt the same value in the simulation boxes. To help in the construction of the simulation boxes, we implemented CLFEP-autoconf.tcl, a tcl script that automatically builds all the structural and configuration files needed to run the sampling simulations of the initial and final states of the systems.

3.7.III. Careful Sampling of the Bound State. As with other free energy approaches, such as umbrella sampling⁷⁷ or eABF,^{7,8,43,51} the use of harmonic wall restraints may be required for small host–guest complexes, with low or moderate affinity. These restraints should be sufficient to keep the guest within the host, thus limiting the sampling of high-energy geometries or even unbound states of the systems. Our tool CLFEP_autoconf.tcl allows setting up a wall-type restraint, located at the distance between the center of masses of the host and the guest, plus a tolerance distance with a default value of 5 Å. This restraint has the sole purpose of maintaining the integrity of the complex. Therefore, the tolerance value should be fixed at a sufficiently large distance to allow sampling all possible bound state conformations. Nonetheless, like in other free energy methods, an adequate sampling of the states of interest is key to obtain an accurate estimate of the free energy change.

3.7.IV. Careful Sampling of the Reference State. In addition to the complex, the unbound state must be carefully sampled. Frequently, unbound states are built using a previously reported structure of the bound state. Hence, we advise to carefully check for specific structural features of the bound state that may not be part of the actual initial states. These features could be large conformational changes, water molecules stabilizing interactions near or in the binding site, changes in the protonation states of the ligand or the host, among others. A relaxation of the unbound systems is thus advised prior sampling the energies for the analysis.

3.7.V. Standard State and Finite-Size Corrections. Both corrections are automatically applied during the evaluation of CL-FEP as described in the [Supporting Information](#). The standard-state correction is a function of the ratio of the concentration in the computed system and the standard state concentration value (1 M).⁷⁸ The self-interaction energy differences between initial and final states is corrected based on the work of Hünenberger et al.⁷⁹ (Supplementary Note). The later correction to finite-size effects assumes a cubic geometry for the simulation boxes. Our tool, CLFEP_autoconf.tcl, assures this feature during the construction of the simulation boxes.

3.7.VI. Prevent the Use of Switching Functions. The default configuration of NAMD for molecular dynamics simulations applies a switching function to the Lennard-Jones potential in order to enhance the energy conservation during the simulations.⁸⁰ This procedure scales down to zero the energetic contributions of van der Waals (VdW) interactions near a predefined cutoff (e.g., 12 Å). However, the direct use of the resulting potential energy values can produce an underestimation of dispersive VdW interactions in the free energy change between the bound and unbound states. Thus, it is advised to avoid the use of switching functions or to re-evaluate the energy of all the sampled frames in the simulations without applying the switching procedure. This feature can affect particularly those systems where dispersive interactions play a key role in complex stability like in benchmark sets (ii) and (iii).

3.7.VII. Double Check out-of-Range Predictions. While these guidelines are valid for any estimation, out-of-range predictions should immediately trigger a deeper analysis of the

sampling. The user should be aware that the affinity of most noncovalent binders lies between ~ -2 kcal/mol for very weak ligands and ~ -15 kcal/mol for the most potent binders.⁸¹ CL-FEP is, in principle, an unbiased estimator, thus values predicted outside that range might originate from shortcomings in the sampling or in the setup of the model system. In such cases, we recommend the following: 1) If the used ρ_{OSR} is larger than 1, to increase the actual sampling and reduce ρ_{OSR} . 2) To adjust the total charge concentration in the simulation to resemble, as close as possible, the experimental conditions. 3) To explore the effect of any restraint applied to the system or, if no restraints were used, consider fixing a boundary potential to keep the sampling within the targeted state. This may be essential for complexes with weak affinity (-5 kcal/mol or less) and exposed binding sites. 4) Check the composition of the systems, as crystallization conditions often differ from the conditions used to determine absolute binding affinities. Thus, the lack or inadequate addition of cofactors and allosteric ligands may affect the calculated values.

The points discussed above are related to critical issues that originate from the construction of the four simulation boxes used to sample the bound and unbound states. In this work, we followed a similar protocol to end-state approaches, such as LIE and MM/(P/G)BSA, where the sampling can be performed with independent simulations of the initial and final states (e.g., the three-average MM/(P/G)BSA, 3A-MM/(P/G)BSA).⁴ In this context, we show that keeping the mass and charge balance among the simulations, CL-FEP produces unbiased predictions of binding free energies, thus validating this approach.

We note that the CL-FEP estimator can also be evaluated with samples of energy differences obtained using alchemical and dual-topology schemes implemented in most molecular dynamics codes. Such protocols can render more robust energy sampling, since the initial and final states are in the same simulation box, thus avoiding an inadequate distribution of particles (water and ions) when different simulation boxes are used to represent the states. Importantly, CL-FEP can also use lists of energies obtained with MD codes such as Amber^{82,83} and GROMACS.^{84,85} Although CLFEP_autoconf currently delivers input files for NAMD runs, we will soon extend it to also provide input files for these MD codes too.

4. CONCLUSION

We developed CL-FEP, which is an end-state solution of the Free Energy Perturbation identity based on the Central Limit Theorem. The approach benefits from a bootstrapping procedure over the energies of the initial and final states of the system transformation. CL-FEP achieves an effective reduction of the energy variance in explicit solvent simulations, which typically prevents the direct end-state evaluation of free energy estimators. Remarkably, the CL-FEP approach shows excellent accuracy estimating the binding free energy of molecular and biomolecular complexes using energy samples from the end states of the binding process. In the same spirit of FEP and BAR methods, the CL-FEP approach can also be used to process a stratified sampling, which would probably enhance the robustness of the final estimate. However, in the present work we show that even within an end-state sampling scheme, CL-FEP delivers very accurate predictions. Furthermore, the implemented bootstrapping procedure makes the evaluation of the convergence and the error of the estimation an intrinsic part of the algorithm.

Further, we developed CLFEP.pl, a tool that delivers free and user-friendly access to our approach. As input data, only the direct energy samples from initial and final states of the system transformation are needed. The code reads natively the output files of NAMD 2.12 (or 2.13),^{41,42} but it can also read inputs in CSV format. CL-FEP provides an attractive alternative to current end-state estimators of free energy changes and enlarges the repertoire of unbiased free energy approaches by delivering accurate estimates of free energy changes, at a reduced computational cost.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.9b00725>.

Thermodynamic and convergence parameters using second order cumulant estimator (C2) and notes on finite-size corrections and enthalpy estimations (PDF)
Input and output files from CL-FEP analysis of case studies (ZIP)

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Notes

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